Sahaj Bhadja

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Summary

Machine Learning Engineer with an MS in CS and 2+ years building **PyTorch** models, **classification** and **recommendation** systems, embedding retrieval, and production **LLM** workflows in **Python**, from feature pipelines to evaluation and deployment.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Advanced NLP, Analysis of Algorithms, Database Systems, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Data Retrieval, Software Engineering, Object-Oriented Programming, Operating Systems

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*
- Built fault-tolerant **LLM orchestration** for an AI education platform with FastAPI workers, explicit job states, and failure isolation, processing 1K+ AI jobs at 3x throughput under concurrent load.
- Built **data pipelines** across PostgreSQL, SQL Server, and cloud storage with incremental loads, validation, and reconciliation for 500+ users, running 5K+ monthly jobs at 99%+ production reliability.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*
- Deployed a **multi-step LLM workflow** for education that generates rubrics, extracts handwritten answers via OCR, grades, and explains feedback across GPT-4o, Claude, and Qwen-VL for 150+ instructors.
- Sustained 90%+ grading consistency with switchable **model routing** and timeout fallback across providers, tracking per-call token cost while scaling to 10K+ daily requests and cutting feedback turnaround 60%.

USC AD Labs **January 2025 – May 2025**
ML Research Assistant *Los Angeles, CA*
- Trained a GPT-based Decision Transformer in **PyTorch** with cross-entropy over discretized actions, AdamW, and cosine decay on features engineered from 100K+ trajectory logs, improving long-horizon planning 18% in simulation.
- Designed a multi-head attention encoder fusing map and agent signals, tracked experiments in **W&B** with Hydra configs, and evaluated against three baselines, improving closed-loop planning stability 15%.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*
- Redesigned **ETL** on AWS Glue and S3 that consolidated user activity logs from multiple sources into analytics-ready formats, boosting pipeline efficiency 68% and cutting storage costs 15%.
- Optimized large-scale API integrations with caching and pagination, improving response times 30%, and partnered **cross-functionally** with Microsoft and GitHub stakeholders in Agile sprints with 20+ automated test suites.

Cyber Security and Data Mining: Detecting Malicious URL **May 2023 – August 2023**
Data Analyst – Research Fellowship *India*
- Engineered 21 behavioral **features** such as IP presence, URL length, and suspicious keywords over a 650K+ URL dataset, building an end-to-end classification pipeline for phishing and malware detection.
- Trained and evaluated Gaussian Naive Bayes, Random Forest, LSTM, Bi-LSTM, GRU, and CNN **classifiers**, comparing model performance to reach 96%+ detection accuracy and reporting actionable phishing and malware insights.

Skills

Languages: Python, SQL, Java, C++, TypeScript
ML & Deep Learning: PyTorch, TensorFlow, scikit-learn, Pandas, Transformers, CNN, LSTM, Random Forest, Offline RL
ML Concepts: Classification, Ranking, Recommendation Systems, Feature Engineering, Model Evaluation, Embeddings, Experiment Tracking
LLMs & Retrieval: LLM Workflows, Prompt Chaining, RAG, FAISS, Pinecone, Claude, GPT-4o, LLaMA-2
Data & Cloud: Spark, Databricks, AWS Glue, PostgreSQL, ETL Pipelines, AWS, GCP, Azure, Docker
Tools: Weights & Biases, Hydra, FastAPI, Git, CI/CD

Projects

Two-Stage Job Recommender | *Python, PyTorch, FAISS, LightGBM, scikit-learn, Pandas*
- Designed a two-stage recommender with **two-tower embedding retrieval** over FAISS and a LightGBM **ranker** on engagement features, evaluated offline with NDCG@k and recall@k and explained with SHAP feature attributions.

Trend Tracker with Fashion Recommendation System Live | *CNN, React.js, Strapi, AWS EC2*
- Built and deployed a CNN-based **recommendation** model for an e-commerce trend tracker on AWS EC2, improving recommendation accuracy from 78% to 92% and lifting engagement across 3M+ users.

Big-Brain Live | *LLaMA-2, FAISS, Pinecone, Embeddings, PostgreSQL*
- Built a semantic retrieval system with transformer **embeddings** over FAISS and Pinecone across 50K+ documents, powering LLaMA-2 Q&A at 95% query accuracy and cutting retrieval latency 40%.